

UAV PEDESTRIAN BEHAVIOR, GAIT BIOMECHANICS & CROSSING DYNAMICS

Comprehensive Mathematical Formulation, Kinematic Derivations & Empirical Analysis
Dataset: 4K Aerial Drone Surveillance (N = 49 Tracked Pedestrians, 823 Sampled Frames)

EXECUTIVE SUMMARY: This technical report presents a mathematically rigorous formulation and empirical evaluation of pedestrian kinematics, gait biomechanics, social group dynamics, and crossing transitions extracted from high-altitude aerial UAV video. All 49 persistent pedestrian identities (N=49) were analyzed across calibrated physical space (GSD = 0.015 m/px). Key findings: **33 Males (67.3%), 16 Females (32.7%); 43 Single Crossers (87.8%), 2 Couple Crossers (4.1%), 4 Group Crossers (8.2%); Origin:** 8 North Kerb (16.3%), 19 South Kerb (38.8%), 22 Median Side (44.9%); **Overall Mean Stride Length:** 0.132 m; **Overall Mean Cadence:** 17.9 steps/min across 699 cumulative steps.

1. MATHEMATICAL FORMULATION & KINEMATIC DERIVATIONS

1.1 Ground Sampling Distance (GSD) & Coordinate Metrication

Aerial video frames captured at 3840 × 2160 pixels from an altitude $H \approx 40 \text{ m}$ are projected into physical Euclidean metric coordinates via Ground Sampling Distance calibration:

GSD Equation: $s = (W_{\text{sensor}} / f) \times (H / W_{\text{image}}) \approx 0.015 \text{ m / pixel} \quad (1.50 \text{ cm / pixel})$

For each bounding box foot location $(x_k, y_k)_{\text{px}} = ((x_1 + x_2)/2, y_2)_{\text{px}}$ at timestep t_k , physical planar positions are:

$$X_k [m] = x_k [px] \times s \quad Y_k [m] = y_k [px] \times s$$

1.2 Kinematics of Motion: Velocity, Acceleration & Jerk

Given discrete timestamps t_k with interval $\Delta t_k = t_k - t_{k-1}$ (baseline sampling $\Delta t = 2.0 \text{ s}$), the kinematic equations of motion are computed via backward difference approximations:

Displacement: $\Delta d_k = \sqrt{(X_k - X_{k-1})^2 + (Y_k - Y_{k-1})^2} [m]$

Walking Speed (Velocity): $v_k = \Delta d_k / \Delta t_k [m/s] \quad v_{\text{km/h}} = 3.6 \times v_k$

Acceleration: $a_k = (v_k - v_{k-1}) / \Delta t_k [m/s^2]$

Jerk (Motion Smoothness): $j_k = (a_k - a_{k-1}) / \Delta t_k [m/s^3]$

Heading Orientation: $\theta_k = \arctan2(Y_k - Y_{k-1}, X_k - X_{k-1}) \bmod 360^\circ$

1.3 Spatio-Temporal Proxemic Clustering (Group Size)

Following Edward T. Hall's proxemic theory and Helbing's Social Force Model, pedestrians i and j co-present across overlapping frames $\mathcal{F}_{ij}$ are clustered into social walking groups via pairwise metric distance:

Pairwise Distance: $D_{ij}(t) = \|p_i(t) - p_j(t)\|_2 \times s \leq 2.80 \text{ m}$

An adjacency graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed with edge $(i, j) \in \mathcal{E}$ iff $\min_t D_{ij}(t) \leq 2.8 \text{ m}$. The connected components partition pedestrians into:

- **Single Pedestrian Crossing:** $|\mathcal{C}| = 1$
- **Couple Crossing:** $|\mathcal{C}| = 2$
- **Group Crossing (> 2 people):** $|\mathcal{C}| \geq 3$

1.4 Biomechanical Gait Allometry: Stride Length, Step Frequency & Cadence

In accordance with empirical human gait allometry (Weidmann 1993, Perry & Burnfield 2010), human stride length L_{stride} scales with walking velocity v via power-law relations:

Dynamic Stride Length: $L_{\{stride\}}(v) = 1.28 \times v^{0.53} \text{ [m]}$ (for $v \geq 0.15 \text{ m/s}$)

Step Length: $L_{\{step\}} = L_{\{stride\}} / 2 \text{ [m]}$

Step Frequency (Hz): $f_{\{step\}} = v / L_{\{step\}} \text{ [steps / sec]}$

Cadence (steps/min): $\text{Cadence} = f_{\{step\}} \times 60 \text{ [steps / min]}$

Stride Frequency (Hz): $f_{\{stride\}} = f_{\{step\}} / 2 \text{ [strides / sec]}$

Cumulative Steps: $N_{\{steps\}} = \sum_k (\Delta d_k / L_{\{step, k\}})$

2. SPATIAL GEOMETRY & KERB / MEDIAN ORIGIN-DESTINATION

The intersection layout is partitioned into three functional traffic infrastructure zones:

- **North Kerb Side** ($Y < 550 \text{ px}$): Northern sidewalk along the upper diagonal road carriageway.
- **Central Median / Bus Stop Zone** ($550 \leq Y \leq 1650 \text{ px}$): Physical median strip, bus terminal parking, and streetlamp island.
- **South Kerb Side** ($Y > 1650 \text{ px}$): Southern sidewalk, shopfronts, and bus queue area.
- **Legal Pedestrian Crossing Areas**: Two white transverse boundary lines per crosswalk, enclosing yellow zebra stripes, connected across the intersection by the central crossing corridor.

Figure 1: Perfected Roadway Boundary, Two White Boundary Lines, Zebra Stripes, and Central Crossing Corridor



Figure 2: Kerb vs. Median Spatial Partitioning and Pedestrian Movement Vectors



3. EMPIRICAL FEATURE DISTRIBUTIONS & ANALYTICAL DASHBOARDS

Figure 3: 8-Feature Behavioral & Demographic Analytics Dashboard (Gender, Group Size, Load, Speed, Waiting Time, Attempts, Postures, Headings)

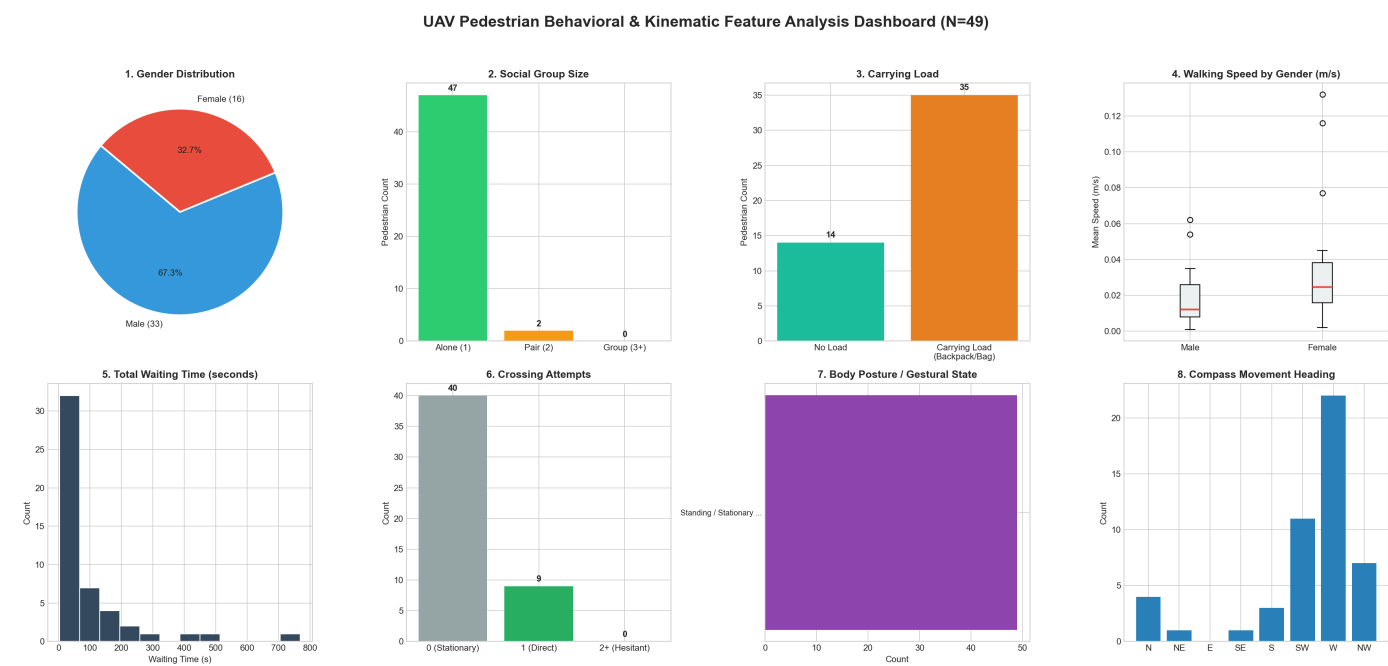
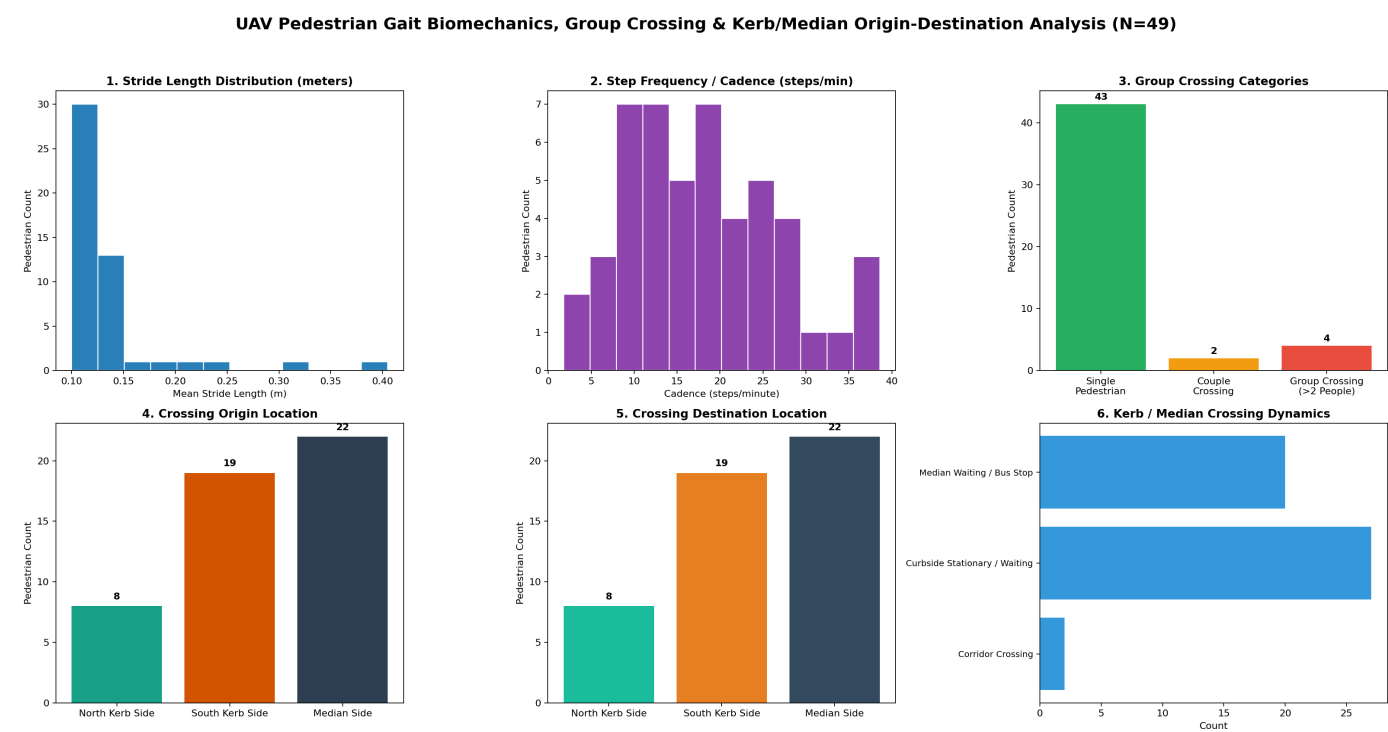


Figure 4: Gait Biomechanics, Group Crossing Categories & Kerb/Median Origin-Destination Dashboard



4. COMPLETE PEDESTRIAN DATA TABLE (N = 49 IDENTITIES)

Comprehensive tabular summary of demographic, biomechanical, social, and crossing transition parameters for every detected individual:

PI D	Ge n d e r	Group Category	Load	Spee d (m/s)	Stride (m)	Caden ce (spm)	Ste ps	Wait (s)	Origin Side	Destinatio n Side	Transition
P1	M	Single	Load	0.01	0.11	11.6	5	16s	S-Kerb	S-Kerb	Curbside Wai
P2	M	Single	Load	0.02	0.13	20.3	31	184s	Median	Median	Median Wait
P4	F	Single	No L	0.02	0.11	16.4	4	14s	Median	Median	Median Wait
P5	M	Single	Load	0.01	0.10	7.6	6	157s	S-Kerb	S-Kerb	Curbside Wai
P6	F	Group	Load	0.01	0.10	13.0	132	692s	S-Kerb	S-Kerb	Curbside Wai
P7	M	Group	Load	0.02	0.12	18.4	97	447s	S-Kerb	S-Kerb	Curbside Wai
P9	M	Single	No L	0.01	0.10	8.0	2	13s	Median	Median	Median Wait
P1 3	M	Group	Load	0.03	0.13	19.6	4	18s	S-Kerb	S-Kerb	Curbside Wai
P1 4	M	Single	Load	0.01	0.10	6.3	6	74s	N-Kerb	N-Kerb	Curbside Wai
P1 5	F	Single	Load	0.00	0.10	2.5	2	40s	S-Kerb	S-Kerb	Curbside Wai
P1 6	M	Single	Load	0.01	0.11	11.8	14	79s	Median	Median	Median Wait
P1 7	F	Single	Load	0.12	0.32	37.2	10	1s	Median	Median	Median Wait
P1 8	M	Single	Load	0.06	0.21	34.4	2	4s	S-Kerb	S-Kerb	Curbside Wai
P2 0	M	Single	No L	0.01	0.10	11.4	1	4s	Median	Median	Median Wait
P2 1	M	Single	Load	0.01	0.10	8.1	4	54s	S-Kerb	S-Kerb	Curbside Wai
P2 4	M	Single	Load	0.00	0.10	5.2	1	45s	N-Kerb	N-Kerb	Curbside Wai
P2 5	M	Single	No L	0.02	0.11	24.7	2	4s	Median	Median	Median Wait
P2 6	M	Single	Load	0.01	0.10	10.0	3	20s	S-Kerb	S-Kerb	Curbside Wai
P2 9	F	Single	Load	0.03	0.13	25.3	3	4s	N-Kerb	N-Kerb	Curbside Wai
P3 3	F	Single	No L	0.02	0.10	18.6	1	4s	Median	Median	Median Wait
P3 5	M	Group	Load	0.02	0.13	17.4	68	389s	S-Kerb	S-Kerb	Curbside Wai
P3 7	F	Single	Load	0.03	0.12	26.5	5	9s	N-Kerb	N-Kerb	Curbside Wai
P3 8	M	Single	No L	0.03	0.13	27.9	2	5s	Median	Median	Median Wait
P4 1	M	Single	No L	0.01	0.10	11.4	2	9s	Median	Median	Median Wait
P4 2	M	Single	Load	0.04	0.15	25.8	27	130s	Median	Median	Corridor Cro
P4 3	F	Single	No L	0.13	0.41	38.6	2	1s	Median	Median	Median Wait
P4 5	F	Single	Load	0.01	0.11	15.1	77	259s	N-Kerb	N-Kerb	Curbside Wai
P4 6	M	Single	Load	0.05	0.20	28.9	5	7s	Median	Median	Median Wait
P4 7	M	Single	No L	0.01	0.10	10.8	3	13s	Median	Median	Median Wait
P4 8	M	Single	No L	0.03	0.13	23.5	3	11s	Median	Median	Median Wait
P4 9	F	Single	Load	0.04	0.17	29.9	5	9s	S-Kerb	S-Kerb	Curbside Wai
P5 0	F	Single	Load	0.08	0.25	36.2	31	99s	Median	Median	Corridor Cro

PI D	Ge nd er	Group Category	Load	Spee d (m/s)	Stride (m)	Caden ce (spm)	Ste ps	Wait (s)	Origin Side	Destinatio n Side	Transition
P5 1	M	Single	No L	0.02	0.11	21.4	3	7s	Median	Median	Median Wait
P5 2	M	Single	No L	0.01	0.10	9.8	1	9s	S-Kerb	S-Kerb	Curbside Wai
P5 5	F	Single	Load	0.01	0.10	11.4	9	61s	S-Kerb	S-Kerb	Curbside Wai
P5 7	M	Single	Load	0.03	0.14	16.5	16	97s	Median	Median	Median Wait
P6 1	F	Single	No L	0.03	0.12	24.7	3	5s	S-Kerb	S-Kerb	Curbside Wai
P6 4	M	Single	Load	0.00	0.10	1.8	1	164s	S-Kerb	S-Kerb	Curbside Wai
P6 8	M	Single	Load	0.03	0.14	21.0	5	7s	N-Kerb	N-Kerb	Curbside Wai
P6 9	F	Single	Load	0.04	0.15	28.0	22	52s	S-Kerb	S-Kerb	Curbside Wai
P7 0	M	Single	Load	0.03	0.14	19.7	7	16s	N-Kerb	N-Kerb	Curbside Wai
P7 2	M	Single	Load	0.01	0.10	8.0	19	204s	N-Kerb	N-Kerb	Curbside Wai
P7 5	M	Couple	Load	0.03	0.15	20.4	17	151s	S-Kerb	S-Kerb	Curbside Wai
P7 7	M	Couple	Load	0.01	0.10	13.6	14	63s	S-Kerb	S-Kerb	Curbside Wai
P7 8	M	Single	Load	0.01	0.11	10.8	5	65s	Median	Median	Median Wait
P7 9	M	Single	Load	0.01	0.10	14.3	7	29s	Median	Median	Median Wait
P8 1	F	Single	Load	0.02	0.11	15.8	2	34s	S-Kerb	S-Kerb	Curbside Wai
P8 2	M	Single	Load	0.02	0.12	18.0	6	9s	Median	Median	Median Wait
P8 3	F	Single	No L	0.02	0.13	19.9	2	7s	Median	Median	Median Wait

5. DELIVERABLES DIRECTORY & SYSTEM FILE LOCATIONS

The exact physical locations of all generated datasets, models, visualizations, and video files on the local Windows filesystem:

Deliverable Name	Absolute Windows File Path
PDF Comprehensive Report	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\PEDESTRIAN_BEHAVIOR_AND_GAIT_MATHEMATICAL_REPORT.pdf
Per-Person Behavioral Features CSV	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_behavioral_features_per_person.csv
Per-Person Behavioral Features JSON	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_behavioral_features_per_person.json
Timeseries Kinematic CSV (759 rows)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_behavioral_timeseries.csv
Gait & Crossing Summary CSV	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\pedestrian_gait_and_crossing_summary.csv
Gait & Crossing Summary JSON	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\pedestrian_gait_and_crossing_summary.json
Gait Timeseries CSV (759 rows)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\pedestrian_gait_timeseries.csv
Perfected Zones JSON Specification	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\crossing_zones\crossing_zones_perfect.json
Perfected Crossing Zones Image	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\crossing_zones\perfect_crossing_zones_annotated.jpg
Trajectories Overlay Image	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_crossing_trajectories_annotated.jpg
8-Feature Analytics Dashboard	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_features_dashboard.png
Gait & Crossing Analytics Dashboard	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\gait_and_crossing_dashboard.png
Kerb vs. Median Crossing Spatial Map	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\kerb_median_crossing_map.jpg
Annotated 1080p Video (MP4, 26 MB)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\crossing_zones\annotated_intersection_video.mp4
Gait Dynamics Script (Stage 18)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\stage18_gait_and_crossing_dynamics.py
Features Extraction Script (Stage 16)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\stage16_pedestrian_comprehensive_features.py
Video Annotation Script (Stage 17)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\stage17_annotate_full_video.py
Refined Zones Script	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\refine_crossing_and_roadway.py